

YAMAGATA, Japan

WMO: 475880

Lat: 38.2556N

Lon: 140.3461E

Elev: 154

StdP: 99.49

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.6	-4.6	-9.1	1.8	1.2	-8.1	1.9	0.2	5.5	3.4	5.1	3.1	1.5	170	0.397

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.7	33.5	24.4	31.9	23.8	30.3	23.0	25.2	31.2	24.6	30.2	24.0	29.0	2.9	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.6	18.8	27.6	23.0	18.1	27.2	22.5	17.5	26.8	78.0	31.4	75.5	30.4	73.1	29.1	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.9	5.2	4.5	-7.9	35.9	1.2	1.2	-8.7	36.8	-9.4	37.5	-10.1	38.2	-10.9	39.0		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-8.4	26.3	1.1	0.5	-9.2	26.7	-9.9	27.0	-10.5	27.3	-11.3	27.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.5	-0.1	0.5	4.2	10.4	16.6	20.6	24.6	25.7	21.2	14.7	8.0	2.6
	DBStd	9.50	2.29	2.54	3.13	3.65	3.07	2.55	2.77	2.82	3.37	3.11	3.27	2.74
	HDD10.0	1112	314	267	183	38	1	0	0	0	0	3	78	229
	HDD18.3	2754	572	500	439	238	69	8	0	0	11	120	311	487
	CDD10.0	2017	0	0	2	51	207	319	452	487	335	147	17	1
	CDD18.3	619	0	0	0	1	17	77	194	229	96	6	0	0
	CDH23.3	4701	0	0	0	18	174	457	1479	1986	567	20	0	0
	CDH26.7	1620	0	0	0	2	33	101	541	787	156	2	0	0
Wind	WSAvg	1.7	1.6	1.7	1.9	2.0	1.9	1.8	1.7	1.7	1.6	1.5	1.6	1.6
Precipitation	PrecAvg	1383	112	78	84	76	87	120	183	155	137	117	109	125
	PrecMax	1628	188	129	143	128	144	247	419	446	293	276	168	268
	PrecMin	1087	67	33	30	11	32	55	74	49	32	27	25	41
	PrecStd	144	31	23	32	32	32	49	77	91	69	61	33	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.7	11.5	18.8	26.0	29.8	31.2	34.8	35.6	32.6	25.7	19.6	13.4
		MCWB	5.1	6.5	10.0	14.2	17.6	20.3	24.6	24.7	23.7	18.7	13.1	9.2
	2%	DB	6.1	8.2	14.9	22.4	27.3	29.2	33.1	34.0	30.3	23.2	17.1	10.5
		MCWB	3.4	4.4	7.9	12.4	16.2	19.7	24.4	24.7	22.9	17.4	12.1	7.3
	5%	DB	4.6	6.3	12.1	19.7	25.3	27.6	31.5	32.6	28.4	21.5	15.2	8.5
		MCWB	2.3	2.9	6.7	11.5	15.6	19.2	23.8	24.3	22.0	16.3	11.2	5.8
	10%	DB	3.3	4.6	9.8	17.5	23.1	26.1	29.8	31.0	26.6	19.9	13.5	6.9
		MCWB	1.5	2.0	5.3	10.5	14.8	18.9	23.1	23.9	21.0	15.3	10.2	4.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	6.8	10.6	17.0	19.6	23.0	25.9	26.2	24.7	20.4	14.8	9.9
		MCDB	8.1	10.1	16.5	20.5	24.9	27.3	32.2	32.5	30.1	23.7	17.4	12.4
	2%	WB	3.8	4.8	8.8	14.2	18.2	21.7	25.1	25.4	23.7	18.7	13.1	7.7
		MCDB	5.6	7.9	13.4	19.8	23.9	25.8	31.0	31.6	28.7	21.3	15.7	10.1
	5%	WB	2.6	3.4	7.3	12.6	17.1	20.9	24.4	24.8	22.8	17.3	11.8	6.1
		MCDB	4.1	5.6	10.9	17.8	22.2	25.1	29.9	30.7	27.2	20.2	14.2	8.0
	10%	WB	1.7	2.3	5.9	11.4	16.1	20.0	23.8	24.3	22.0	16.1	10.5	4.9
		MCDB	3.1	4.3	9.5	16.2	21.1	24.5	28.6	29.6	25.6	18.9	13.1	6.6
Mean Daily Temperature Range	5% DB	MDBR	5.6	6.7	8.6	10.8	11.0	9.4	8.4	8.7	8.4	8.9	8.1	6.1
		MCDBR	7.5	9.4	13.4	15.1	15.2	12.3	11.1	10.9	10.4	10.4	10.9	8.9
	5% WB	MCWBR	5.2	6.2	7.8	7.4	6.3	5.0	4.1	3.7	4.4	5.3	6.6	6.1
		MCWBR	7.3	9.1	12.0	12.5	11.6	9.4	10.0	9.9	9.3	8.7	9.8	8.6
Clear-Sky Solar Irradiance	taub	taub	0.380	0.421	0.489	0.531	0.545	0.544	0.553	0.539	0.472	0.430	0.403	0.387
		taud	1.994	1.891	1.842	1.832	1.853	1.934	1.976	2.011	2.164	2.183	2.158	2.084
	Ebn at Noon	Ebn at Noon	752	773	762	759	757	758	746	745	775	766	731	716
		Edh at Noon	129	164	191	204	204	188	179	169	137	122	110	111
All-Sky Solar Radiation	RadAvg	RadAvg	1.74	2.52	3.51	4.50	5.04	4.88	4.36	4.45	3.64	2.86	1.99	1.48
	RadStd	RadStd	0.14	0.18	0.30	0.45	0.52	0.33	0.66	0.53	0.28	0.26	0.14	0.14

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	+0.33	N/A	N/A	N/A	N/A

Nomenclature: See separate page